

Circuit A: (Series Connection)

1. Total Resistance  $R_{Total} = R_1 + R_2 + R_3$

$$= 20\Omega + 30\Omega + 50\Omega$$
$$= 100\Omega$$

2. Total Current  $I = \frac{V}{R}$

$$= \frac{125V}{100\Omega}$$

$$= 1.25 \text{ Amperes}$$

3. Current through each resistor:

1.25 Amperes through each resistor

4. Voltage Drop through each resistor:

$$V_R = I \times R$$

$$\begin{aligned} \text{a.) } &= 20 \times 1.25 \\ &= 25V \\ \text{b.) } &= 30 \times 1.25 \\ &= 37.5V \\ \text{c.) } &= 50 \times 1.25 \\ &= 62.5V \end{aligned}$$

5. Power Dissipated through each resistor:

$$P = V \times I \quad (\text{Watts Law})$$

$$\begin{aligned} \text{a.) } &25 \times 1.25 \\ &= 31.25W \\ \text{b.) } &37.5 \times 1.25 \\ &= 46.875W \\ \text{c.) } &62.5 \times 1.25 \\ &= 78.125W \end{aligned}$$